SEMA®

SERO PumpSystems

Self-priming Side Channel Pump SEMA®.

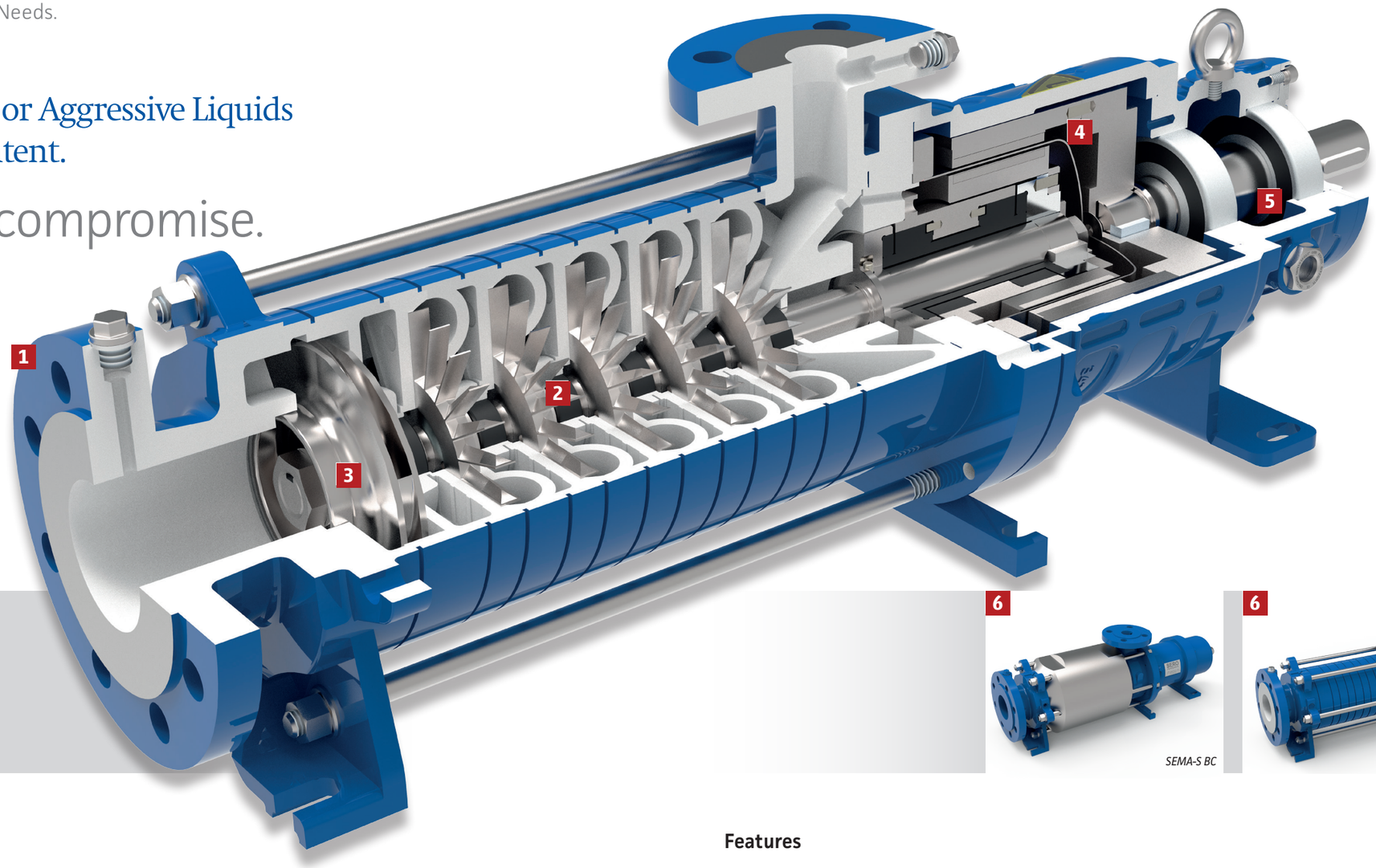
Sealless with Magnetic Drive.

- Gas Handling
- Low NPSH requirements (SEMA-S)
- Leak Proof
- Modular Design with many Standardized Parts

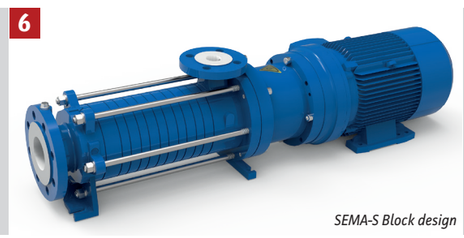
SEMA®

Save handling of Toxic or Aggressive Liquids also with High Gas Content.

SEMA® – No compromise.



SEMA-S BC



SEMA-S Block design

Construction

SEMA 110–660	
Nozzle position	suction: axial discharge: radial
Flanges	accord. EN 1092 PN 40 or ANSI 300 lbs as an option
Shaft seal	sealless due to magnetic coupling
Bearing	Combined medium lubricated axial-radial bearing in SiC or carbon and medium lubricated carbon bearings in every stage, plus two heavy atmospheric and maintenance-free single-row ball bearings greased for bearing lifetime.
Direction of rotation	counterclockwise (seen from drive-end) for SEMA-S and SEMA-L, clockwise for SEMA-Z

Operating data

SEMA 110–660		
Flow rates	up to 36 m^3/h up to 42 m^3/h	(1450 1/min) (1750 1/min)
Heads	up to 350 m up to 550 m	(1450 1/min) (1750 1/min)
Speed	900 - 1800 1/min	
Temperature	-60 °C up to +220 °C	
Nom. Pressure	max. 40 bar (depending on mech. seal)	
NPSH	starts at 0,35 m starts at 1 m	(SEMA-S) (SEMA-L/Z)

Features

1 Robust, compact castings in various designs.

2 Open star impellers with automatic hydraulic axial balance. Free of axial thrust.

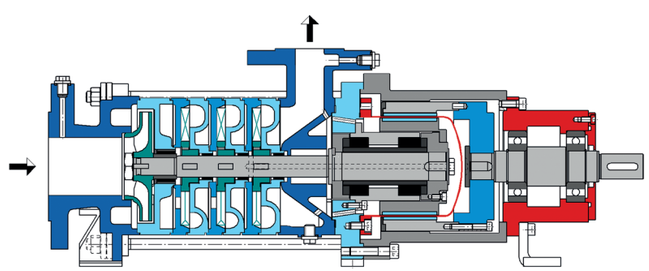
3 SEMA-S with an additional NPSH-impeller guarantees extremely low NPSH values (axial inlet).

4 Permanent magnetic coupling with completely encapsulated inner rotor and oversized SiC drive-end sleeve bearing.

5 The heavy atmospheric ball bearings with lifetime grease-filling are maintenance-free. Oil lubrication as an option.

6 Specific models for special applications are available:

- BC (Barrel casing for elevated environmental and operational safety)
- B (Close-coupled block design for limited space and ideal alignment)
- TT (Low temperature option for media down to -60° C)
- K (High temperature option for media up to +220° C)



SERO PumpSystems

seroweb.com
info@seroweb.de
sales@seroweb.de

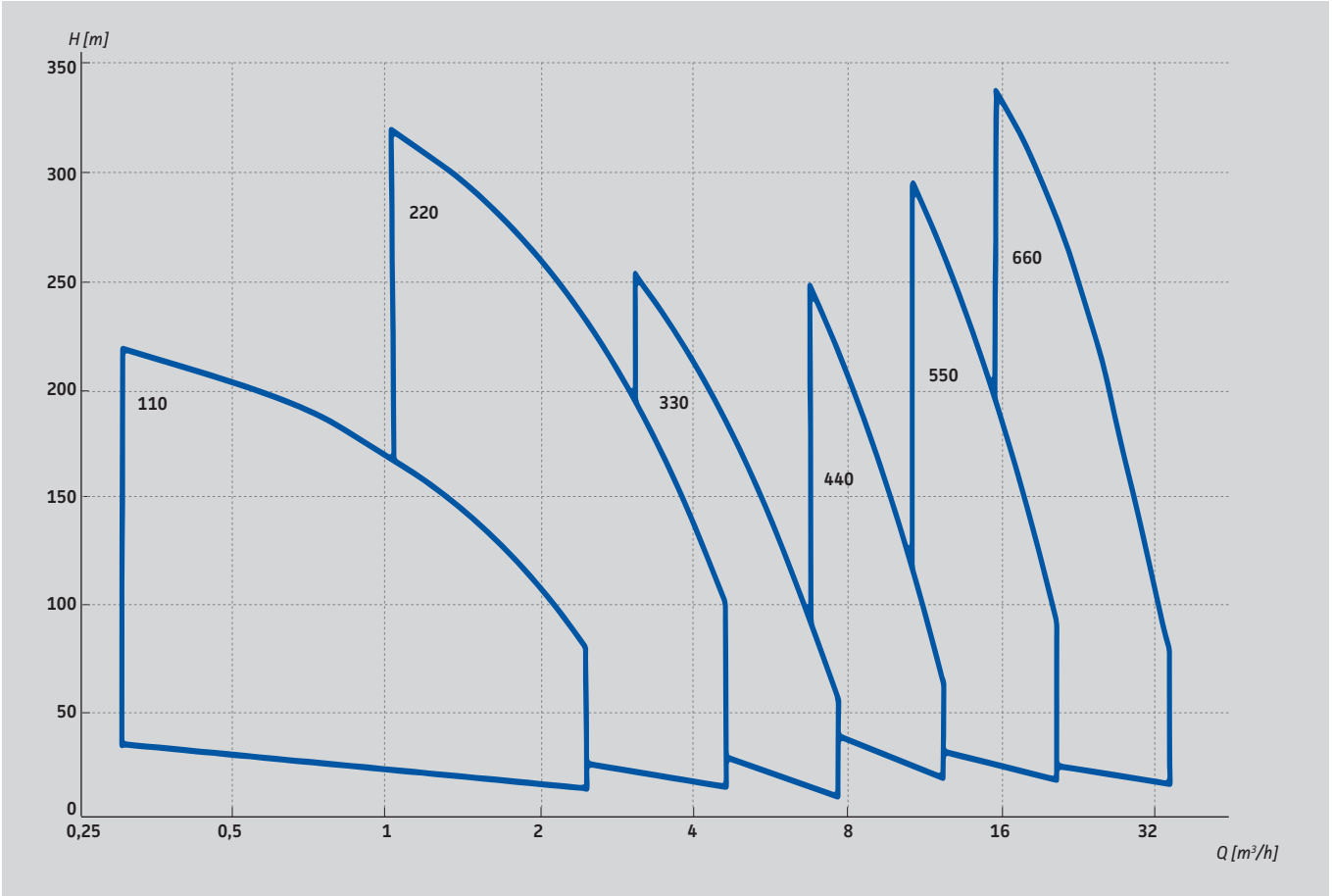
SERO PumpSystems GmbH
Industriestr. 31
74909 Meckesheim/Germany
Phone: +49 6226 9201-0
Fax: +49 6226 9201-55

Material specification

	cast iron (12)	ductile iron (62)	stainless steel (32)
Casings	GJS 400-18	GJS 400-18	1.4408
Inner stage casings	GJL 200	GJS 400-15	1.4470
Shaft	1.4021	1.4021	1.4571
Impeller	1.4059	1.4059	1.4581
Bearing bushing	Carbon	Carbon	Carbon
Sleeve bearing (mag drive)	S2 (SiC-SiC) I S4 (SiC-Carbon)		
Split can	1.4571 [stainless steel] I 2.4610 [Hastelloy C4] I CFK [carbon fiber reinforced plastic]		
Flange (split can)	1.4571 [stainless steel] I 2.4610 [Hastelloy C4] I 1.4462 [duplex steel]		

[Subject to modifications]

Performance range 1450 min-1



07.01.04 | 18 E.J.